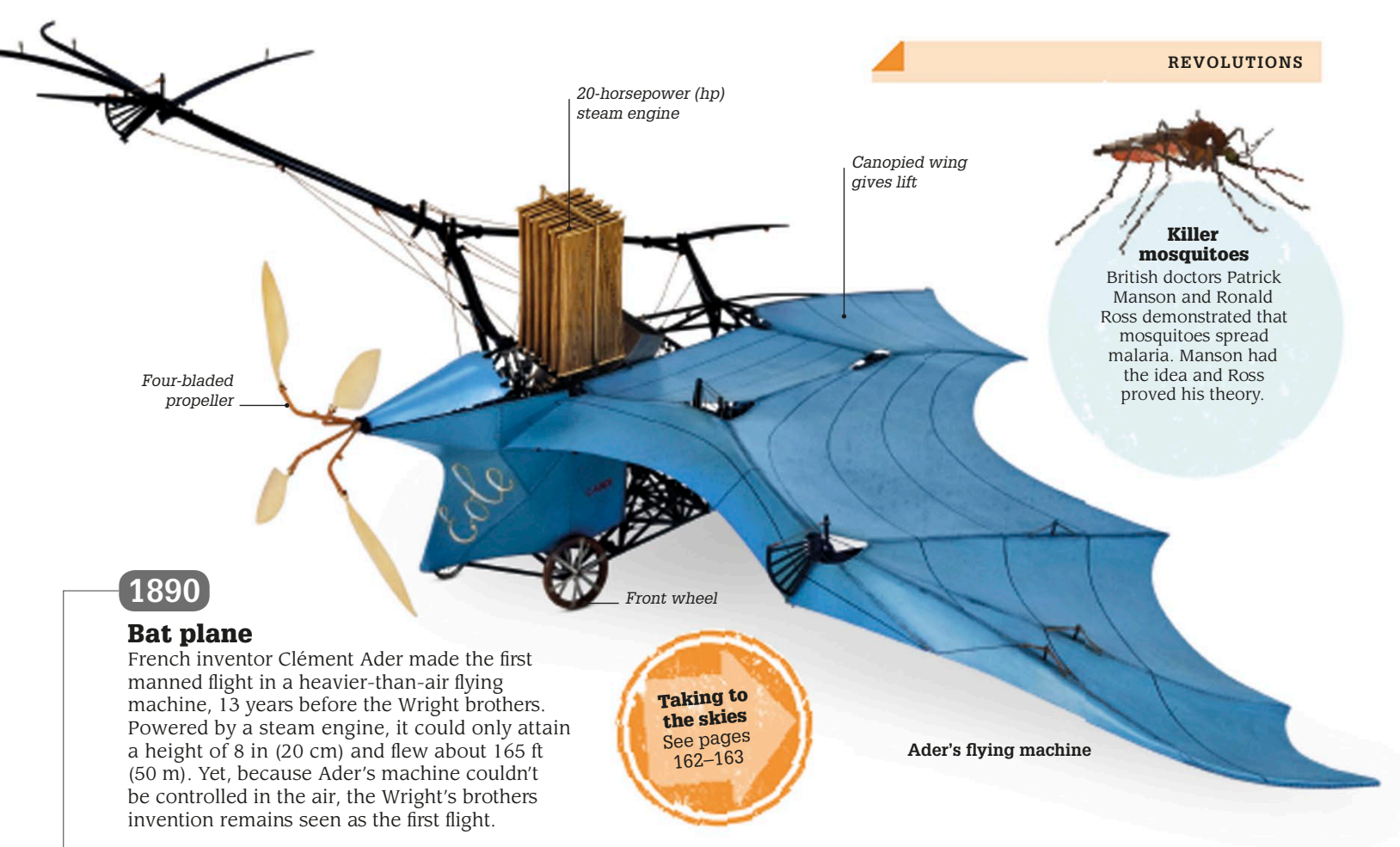




1890

### Bat plane

French inventor Clément Ader made the first manned flight in a heavier-than-air flying machine, 13 years before the Wright brothers. Powered by a steam engine, it could only attain a height of 8 in (20 cm) and flew about 165 ft (50 m). Yet, because Ader's machine couldn't be controlled in the air, the Wright's brothers invention remains seen as the first flight.

**Taking to the skies**  
See pages 162–163

Ader's flying machine

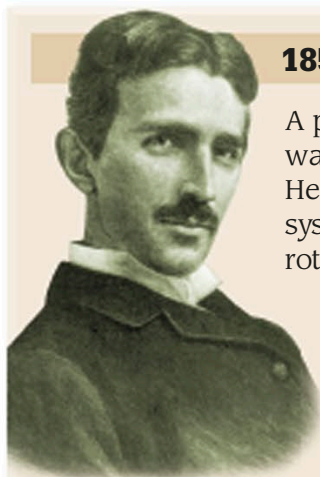


### Killer mosquitoes

British doctors Patrick Manson and Ronald Ross demonstrated that mosquitoes spread malaria. Manson had the idea and Ross proved his theory.

1890

1895



1856–1943

### NIKOLA TESLA

A pioneering figure in the field of electricity, Nikola Tesla was born in Serbia and emigrated to the US in 1884. He helped develop the alternating-current (AC) electrical system (see p.140) widely used today, discovered the rotating magnetic field, and invented the induction motor.

1. The AC supply is connected to the stator.
2. The AC current causes the stator coils to produce a rotating magnetic field, which creates a magnetic field in the rotor coils.
3. The magnetic fields of the stator and rotor coils oppose one another, turning the rotor coil (here hidden from view), which turns a shaft.



Tesla's induction motor

### Induction motor

In Tesla's induction motor, alternating current supplied to a fixed coil (the stator) creates a magnetic field that turns another coil called the rotor, which turns a shaft. Induction motors are used to power large industrial machines, as well as household appliances such as refrigerators, hairdryers, and washing machines.

The noble gas argon was first isolated in 1894. It makes up 0.94 percent of Earth's atmosphere.

